

HR WORKFORCE ANALYTICS

Employee Attrition & HR Analytics SQL Queries

A complete set of 30 SQL queries used to explore, validate, and analyze the employee attrition dataset — from basic aggregations to window functions and composite risk scoring.

30

QUERIES

4,327

EMPLOYEE RECORDS

6

CORRECTIONS APPLIED

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1 Total Employees

AGGREGATION

```
SELECT COUNT(*) AS Total_employee
FROM employee_data;
```

2 Overall Attrition Rate

AGGREGATION

```
SELECT ROUND(SUM(CASE WHEN
Attrition='Yes'
THEN 1 ELSE 0
END)*100.0/COUNT(*),2)
AS Attrition_rate
FROM employee_data;
```

3 Department-wise Attrition Rate

AGGREGATION

```
SELECT Department,
COUNT(*) AS Total_Employee,
SUM(CASE WHEN Attrition='Yes'
THEN 1 ELSE 0 END) AS
AttritionCount,
ROUND(SUM(CASE WHEN Attrition='Yes'
THEN 1 ELSE 0
END)*100.0/COUNT(*),2)
AS AttritionRate
FROM employee_data
GROUP BY Department
ORDER BY Total_Employee DESC;
```

4 Average Salary by Department

AGGREGATION

```
SELECT Department,
ROUND(AVG(MonthlyIncome),2) AS
AverageSalary
FROM employee_data
GROUP BY Department
ORDER BY AverageSalary DESC;
```

5 Top 10 Highest Paid Employees

SORT & LIMIT

```
SELECT EmployeeID, EducationField,
Department, MonthlyIncome
FROM employee_data
ORDER BY MonthlyIncome DESC
LIMIT 10;
```

6 Gender Distribution

AGGREGATION

```
SELECT Gender,
COUNT(*) AS Employee
FROM employee_data
GROUP BY Gender;
```

7 Job Role with Highest Attrition

AGGREGATION

```
SELECT JobRole,
       COUNT(*) AS HighestAttrition
FROM employee_data
WHERE Attrition = 'Yes'
GROUP BY JobRole
ORDER BY HighestAttrition DESC;
```

8 Average Experience by Department

AGGREGATION

```
SELECT Department,
       ROUND(AVG(TotalWorkingYears),2)
       AS AverageExperience
FROM employee_data
GROUP BY Department
ORDER BY AverageExperience DESC;
```

9 Employees by Education Level

AGGREGATION

```
SELECT Education_Label AS
EducationLevel,
       COUNT(*) AS Employees
FROM employee_data
GROUP BY EducationLevel
ORDER BY Employees DESC;
```

10 Salary Hike Analysis

AGGREGATION

```
SELECT Department,
       ROUND(AVG(PercentSalaryHike),2)
       AS AvgSalaryHike
FROM employee_data
GROUP BY Department
ORDER BY AvgSalaryHike;
```

11 Work-Life Balance vs Attrition

AGGREGATION

```
SELECT WorkLifeBalance_Label,
       COUNT(*) AS Employee,
       SUM(CASE WHEN Attrition='Yes'
                THEN 1 ELSE 0 END) AS Attrition
FROM employee_data
GROUP BY WorkLifeBalance_Label;
```

12 Job Satisfaction vs Attrition

AGGREGATION

```
SELECT JobSatisfaction_Label,
       SUM(CASE WHEN Attrition='Yes'
                THEN 1 ELSE 0 END) AS Attrition,
       COUNT(*) AS Employee
FROM employee_data
GROUP BY JobSatisfaction_Label;
```

13 Performance Rating Distribution

AGGREGATION

```
SELECT PerformanceRating_Label
      AS PerformanceRating,
      COUNT(*) AS Employee
FROM employee_data
GROUP BY PerformanceRating_Label;
```

14 Employees With 10+ Yrs Experience

FILTERING

```
SELECT EmployeeID, TotalWorkingYears
FROM employee_data
WHERE TotalWorkingYears > 10;
```

15 Average Income by Job Role

AGGREGATION

```
SELECT JobRole,
      ROUND(AVG(MonthlyIncome),2) AS
AverageIncome
FROM employee_data
GROUP BY JobRole;
```

16 Dept. with Highest Job Satisfaction

AGGREGATION + LIMIT

```
SELECT Department,
      ROUND(AVG(JobSatisfaction),2)
      AS AverageJobSatisfaction
FROM employee_data
GROUP BY Department
ORDER BY AverageJobSatisfaction DESC
LIMIT 1;
```

17 Age Group Analysis

CASE LOGIC

```
SELECT CASE
      WHEN Age < 25 THEN '18-24'
      WHEN Age BETWEEN 25 AND 35 THEN
'25-34'
      WHEN Age BETWEEN 35 AND 45 THEN
'35-44'
      WHEN Age BETWEEN 45 AND 55 THEN
'45-54'
      ELSE '55+'
      END AS AgeGroup,
      COUNT(*) AS Employee
FROM employee_data
GROUP BY AgeGroup;
```

18 Employees Never Promoted

FILTERING

```
SELECT COUNT(*) AS Employee
FROM employee_data
WHERE YearsSinceLastPromotion = 0;
```

19 Average Years at Company

AGGREGATION

```
SELECT Department,
       ROUND(AVG(YearsAtCompany),2)
       AS AverageYearsAtCompany
FROM employee_data
GROUP BY Department;
```

20 Attrition by Business Travel

AGGREGATION

```
SELECT BusinessTravel,
       COUNT(*) AS Employee,
       SUM(CASE WHEN Attrition='Yes'
              THEN 1 ELSE 0 END) AS Attrition
FROM employee_data
GROUP BY BusinessTravel;
```

21 Top 5 Departments by Salary Spend

SORT & LIMIT

```
SELECT Department,
       SUM(MonthlyIncome) AS Salary
FROM employee_data
GROUP BY Department
ORDER BY Salary DESC
LIMIT 5;
```

22 Rank Employees by Salary

WINDOW FUNCTION

```
SELECT EmployeeID, Department,
       MonthlyIncome,
       RANK() OVER (
         ORDER BY MonthlyIncome DESC) AS
       SalaryRank
FROM employee_data;
```

23 Dense Rank Salary Within Dept.

WINDOW FUNCTION

```
SELECT EmployeeID, Department,
       MonthlyIncome,
       DENSE_RANK() OVER (
         PARTITION BY Department
         ORDER BY MonthlyIncome DESC) AS
       DenseRank
FROM employee_data;
```

24 Running Total of Salary

WINDOW FUNCTION

```
SELECT EmployeeID, MonthlyIncome,
       SUM(MonthlyIncome) OVER (
         ORDER BY EmployeeID) AS
       RunningSalary
FROM employee_data;
```

25 Highest Paid Employee per Dept.

CTE + WINDOW FN

```
WITH CTE AS (  
  SELECT *,  
    ROW_NUMBER() OVER (  
      PARTITION BY Department  
      ORDER BY MonthlyIncome DESC) AS  
  RN  
  FROM employee_data  
)  
SELECT * FROM CTE WHERE RN = 1;
```

26 Salary vs. Company Average

SUBQUERY

```
SELECT EmployeeID, MonthlyIncome,  
  CASE WHEN MonthlyIncome >  
    (SELECT AVG(MonthlyIncome)  
     FROM employee_data)  
  THEN 'Above Average'  
  ELSE 'Below Average'  
  END AS SalaryStatus  
FROM employee_data;
```

27 Highest Experience per Dept.

AGGREGATION

```
SELECT Department,  
  MAX(TotalWorkingYears) AS  
  HighestExperience  
FROM employee_data  
GROUP BY Department;
```

28 Average Training by Department

AGGREGATION

```
SELECT Department,  
  ROUND(AVG(TrainingTimesLastYear), 2)  
  AS AvgTraining  
FROM employee_data  
GROUP BY Department;
```

29 Attrition % by Marital Status

AGGREGATION

```
SELECT MaritalStatus,  
  ROUND(SUM(CASE WHEN Attrition='Yes'  
    THEN 1 ELSE 0 END) * 100.0  
    / COUNT(*), 2) AS  
  AttritionPercentage  
FROM employee_data  
GROUP BY MaritalStatus;
```

30 HR Risk Score (Advanced CASE)

CASE LOGIC

```
SELECT EmployeeID, Department, Age,  
  MonthlyIncome, YearsAtCompany,  
  JobSatisfaction, WorkLifeBalance,  
  CASE  
    WHEN JobSatisfaction <= 2  
      AND YearsAtCompany < 3  
      AND WorkLifeBalance <= 2  
    THEN 'High Risk'  
    WHEN JobSatisfaction <= 2  
      OR WorkLifeBalance <= 2  
    THEN 'Medium Risk'  
    ELSE 'Low Risk'  
  END AS HRRiskScore  
FROM employee_data;
```